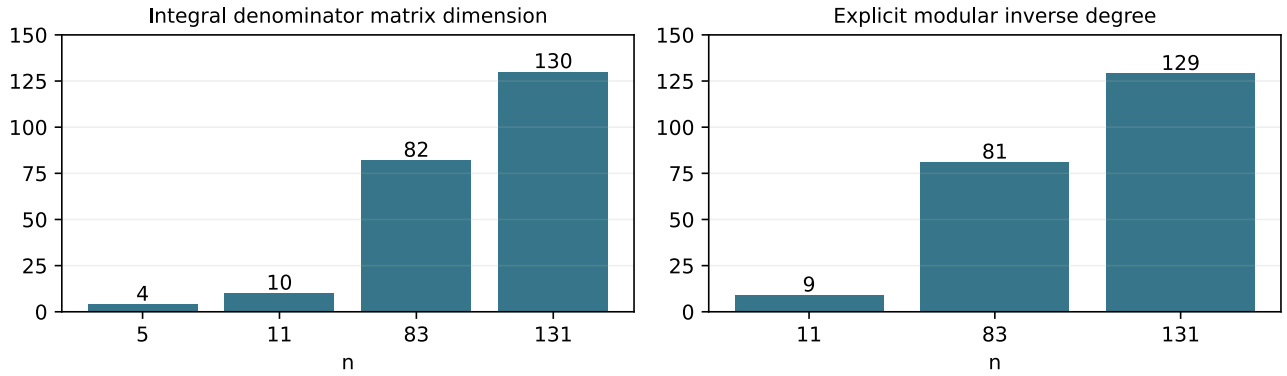
R42: exact larger CM-block recovery

Observations from three native computations and four accepted Lean workflows. No independent evidence verdict, timing comparison or IC speed claim.



integer42-01 and global42-01: complete exact certificates; sizes are not IC costs.

Complete finite calculations

The six fixed cases retain the actual target moduli. The positive n11, n83, n131 cases admit explicit denominator inverses of degrees 9, 81, 129. The n83 and n131 target moduli have 82 and 130 bits. Large primality remains a retained input assumption; integer unit identities need no such assumption. The three controls retain global nonunit obstructions.

For the actual CM exponents, set u=T^i, v=T^j and D=2(v−u)(2u²+3uv+2v²). The opposite quadratic annihilator factors are preserved over the integers, for both twist signs. Complete coefficient certificates give Dred·A=1+Phi·Q+r·H, hence conditional recovery on the stated primitive-character CM block.

Exact integral denominator obstruction

n	Matrix dimension	Least scalar d for this denominator map
5	4	22
11	10	506
83	82	12302938186694
131	130	10095126166020618049226

All original matrix columns, exact Bareiss divisions, adjugate solves and primitive coordinate witnesses are checked natively. A second norm formula checks the determinants. Each measured least scalar matches the prospective Lucas-sequence formula at its fixed n; a universal theorem remains open. The three positive target cases have explicit unit certificates for d.

What is proved, and what is still missing

The integer matrix M is nonsingular and satisfies M·A=d·e0 with primitive coordinates (d,A). Any other integral solve M·z=h·e0 has d·z=h·A. A weighted primitive identity then implies d divides h. This proves minimality for this fixed denominator multiplication lattice. It says nothing about minimality among all geometric maps or all quadratic Bezout choices.

Lean evidence

54 audited declarations across four accepted workflows: 31 base universal and literal unit facts; 15 actual larger-case polynomial identities and recovery corollaries; one generic primitive divisibility bridge; seven actual primitive and target-unit identities. Zero failing or trusted declarations. No native evaluation. Pinned Lean 4.34.1 and mathlib d13f23b; one job, 4096 MiB heap, 1200-second compile/audit caps. The concrete matrix determinant is native rather than formalized. Independent semantic review is pending.

Retained failures and custody

One large-polynomial attempt stopped before compilation while waiting for the base audit receipt. One least-scalar compile rejected a missing finite-sum import; its repair used a fresh attempt. Both are retained. No accepted producer reran and no cap increased. All nine attempt directories are indexed in results.json. Raw artifact byte verification is separate from mathematical replay. Native build/run caps remain 4 GiB virtual memory and 1200 seconds.

Next construction work

Prove the universal denominator formula with exact exceptional hypotheses. Compose the compact integral inverse with the opposite-CM annihilator factors, then bind those operators to actual proper source and target maps. R41 already evaluates two nonzero source characters on an independent collector class; R42 adds larger arithmetic certificates.

The proper level77 map, chosen Koblitz model isogeny, full quotient saturation, polarized embedding, node187 coupled divisors, precision615 source jets, complete prescribed spaces and degree460/367 divisors remain separate open obligations. There is no evaluated larger-n point transfer, IC relation rank, unknown-scalar recovery or speedup here.

The retained R15/R18/R39/R41 notes informed the method. The primary Molin–Page endpoint timed out; no new modularity theorem is asserted. Sources, proof maps, all logs, and full coefficients are linked by REPORT.md, results.json and the immutable artifact manifest.